

EEN115 Final Report

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1 Abstract

In optical communication networks, the routing, modulation format, and spectrum assignment plays an essential role in transferring user information with high-capacity and low delay. The current methods to allocate spectrum suffer from high Number of Cuts (NoC), leading to low utilization and waste bandwidth. To address this issue, a NoC-aware algorithm has been proposed and experimented in two topology networks. In addition, the result of a twining group has been involved to compare with our proposed method.

2 Introduction

Optical communication networks must efficiently allocate limited spectrum resources to support communication demands between different nodes. In this project, we studied the Routing, Modulation format, and Spectrum Assignment (RMSA) problem in elastic optical networks. The input to the problem consists of a physical network topology as seen in Fig. 1, Fig 2, and a traffic demand matrix. The topology describes the set of nodes and fiber links connecting them, along with the distances of the links. The traffic matrix specifies the communication demand between each source to destination pair in the network. Each fiber link is assumed to have 320 frequency slot units (FSUs), with each slot having a width of 12.5 GHz.

The objective of the RMSA problem is to establish lightpaths for all traffic requests while satisfying spectrum constraints such as spectrum continuity, spectrum contiguity, and spectrum clash. In this project, the optimization objective assigned to our group is to minimize spectrum fragmentation, which is measured using the Number of Cuts (NoC) metrics. Lower fragmentation improves spectrum utilization and increases the probability that future requests can be successfully allocated.

To evaluate different approaches, we implement both a benchmark algorithm and a proposed RMSA algorithm. The benchmark algorithm follows a fixed shortest-path routing strategy combined with the First-Fit spectrum assignment method. For the proposed best-fit Noc-aware algorithm, the objective of minimizing Noc is also factored in.

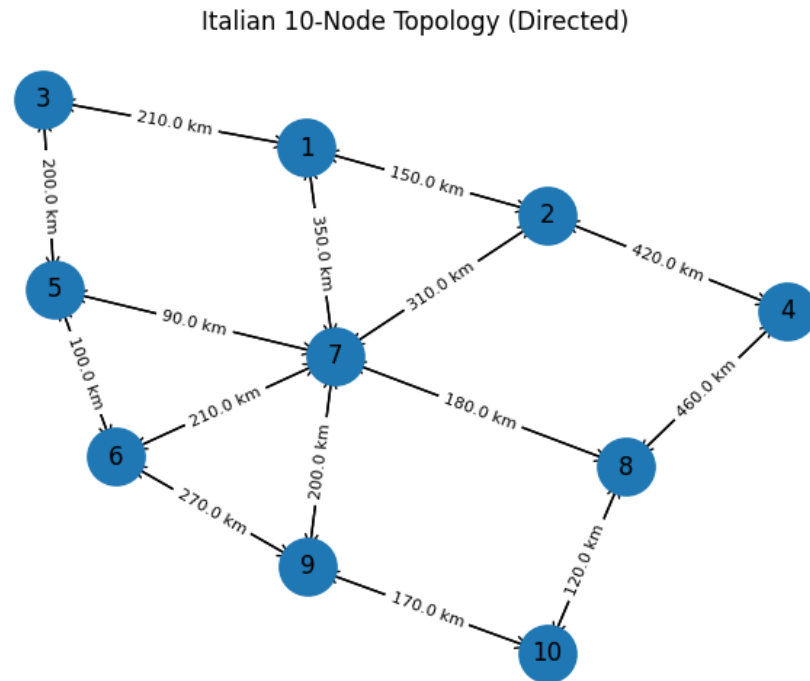


Fig 1. The Italian 10-node topology

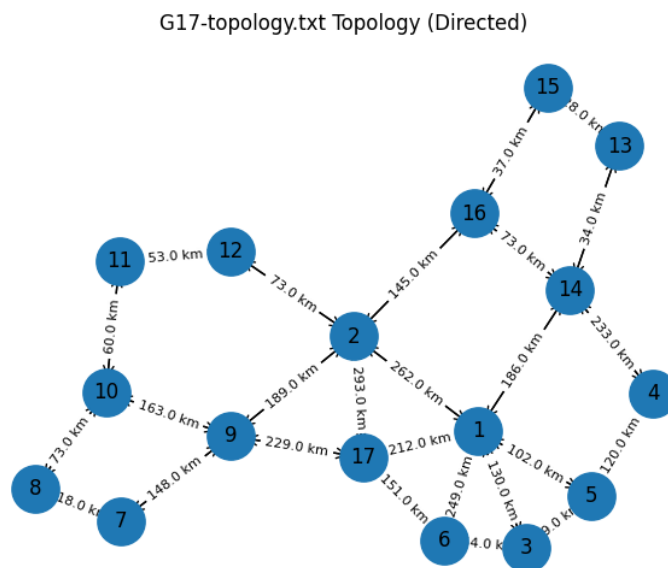


Fig 2. The Germany 17-node topology

3 Benchmark

The benchmark algorithm is a greedy strategy based on the First-Fit spectrum assignment rule. First, all traffic requests are sorted according to their demand, either in ascending or descending order. Then, for each request, the shortest path between the source and destination nodes is selected for routing. Based

on the length of the selected path, the highest-capacity modulation format that satisfies the transmission reach constraint is chosen. After determining the modulation format, the required number of spectrum slots is calculated. The algorithm then scans the spectrum sequentially from the lowest slot index to find the first contiguous block of available slots that satisfies the spectrum of continuity and spectrum contiguity constraints along the entire path. If such a block of slots is found, the slots are allocated to the request; otherwise, the request is blocked.

4 Designed Algorithm

The custom RMSA algorithm allocates traffic requests by selecting the routing path and spectrum slots that minimize spectrum fragmentation in the network. When a request arrives, the algorithm first computes up to five shortest paths between the source and destination. For each path, a feasible modulation format is selected based on the transmission distance, which determines the number of spectrum slots required. If the requested bitrate exceeds the modulation capacity, the request is divided into smaller chunks and allocated sequentially.

For each chunk, the algorithm searches for all possible contiguous free slot blocks along each candidate's path. Each candidate's allocation is temporarily applied, and the resulting network fragmentation is measured using the total Number of Cuts (NoC), which counts the transitions between occupied and free slots across all links. The temporary allocation is then rolled back, and the algorithm selects the candidate that produces the lowest NoC. The best allocation is committed, and the process repeats until the request is fully served, or no feasible allocation exists.

This approach aims to reduce spectrum fragmentation and preserve larger contiguous slot blocks, which can improve future allocation opportunities and potentially lower blocking probability. However, because the algorithm evaluates multiple paths and slot positions with temporary allocations, it has a higher computational cost than simple heuristics such as shortest-path first-fit (Benchmark Algorithm).

5 Performance Evaluation

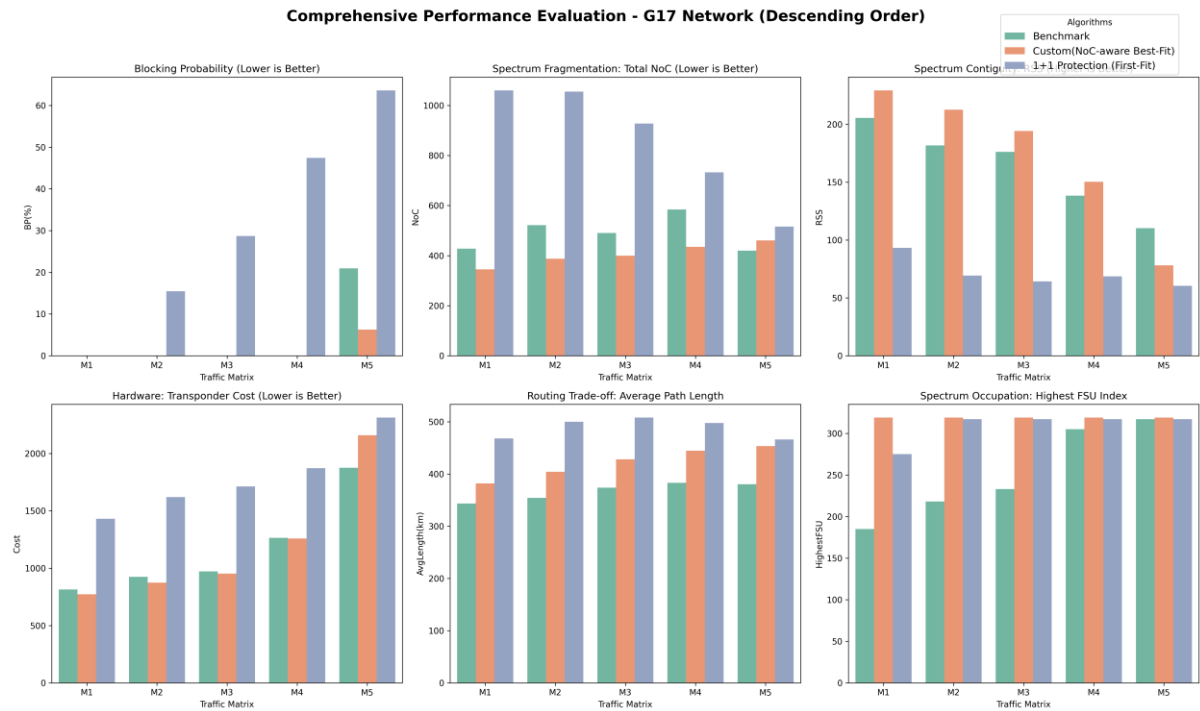


Fig 3. The algorithm is executed on the Germany 17-node network.

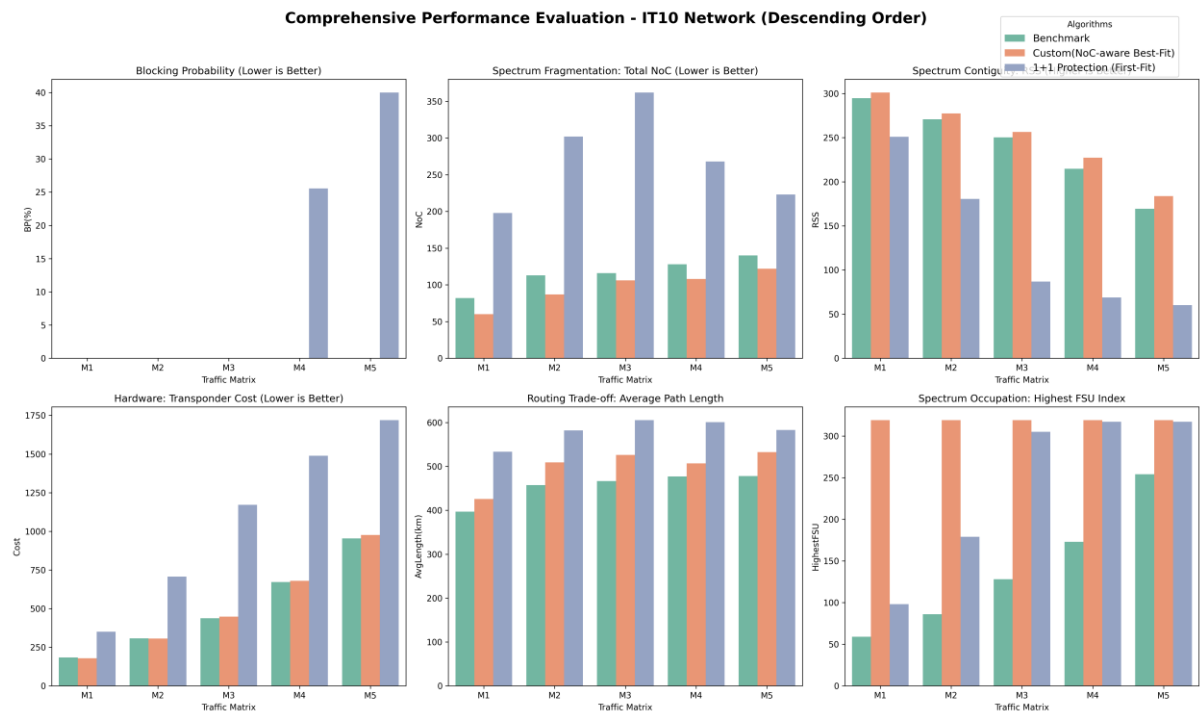


Fig 4. The algorithm is executed on the Italy 10-node network.

5.1 Blocking Probability

Blocking Probability (BP) serves as the primary indicator of network capacity. As vividly illustrated in the top-left subplots for both G17 and IT10, our Custom algorithm vastly outperforms the Benchmark. By exploring alternative paths ($K=5$) to avoid bottleneck links, the Custom algorithm achieves near-zero blocking in low-to-medium loads and reduces the BP in the extreme G17 M5 scenario from 20.96% (Benchmark) down to just 6.25%.

Conversely, the 1+1 Protection scheme triggers a catastrophic spike in blocking probability. In G17 M5, its BP reaches a staggering 63.6%. This is physically intuitive: 1+1 protection requires allocating spectrum on two completely disjoint paths simultaneously. Because it consumes exactly twice the bandwidth for a single demand, the network reaches its capacity limit much faster, starving subsequent connections.

5.2 Number of Cuts (NoC)

As the primary objective of Group 5, minimizing the Number of Cuts (NoC) is where our Custom algorithm truly shines. The charts demonstrate that the Custom algorithm consistently maintains the lowest NoC (e.g., reducing NoC from 584 to 435 in G17 M4 compared to the Benchmark), successfully keeping the spectrum highly contiguous.

On the other hand, 1+1 Protection wreaks havoc on spectrum fragmentation. Its NoC is frequently more than double that of the Benchmark (exceeding 1000 cuts in G17 M1/M2), and its Root of Sum of Squares (RSS) is the lowest across all scenarios. Finding continuous slots on a secondary, often sub-optimal, disjoint path leaves the network's frequency grid heavily fragmented.

5.3 Average length of Established path and Highest FSU Index

To understand why the algorithms perform differently, we must examine the "Average Path Length" and "Highest FSU Index" charts.

The Benchmark maintains the shortest path lengths but artificially blocks traffic early (stopping around FSU index 200-250 in early matrices) because the shortest paths become hopelessly congested.

The Custom Algorithm intentionally trades physical distance for spectrum availability. By routing traffic over slightly longer paths (higher AvgLength), it successfully utilizes the entire spectrum grid (hitting the maximum FSU index of 319 rapidly), thereby accommodating far more traffic.

1+1 Protection forces the longest average paths. Because the primary shortest path is occupied, the

algorithm is forced to route the backup path over the periphery of the network, significantly inflating the average transmission distance.

5.4 Cost of Transponder

The bottom-left chart mapping Transponder Cost reveals the financial impact of routing decisions. Our Custom algorithm incurs a moderately higher hardware cost than the Benchmark. Because it utilizes longer backup paths, optical signals degrade more, forcing the algorithm to step down to robust but lower-capacity modulation formats (e.g., from DP-16QAM to SC-DP-QPSK). Lower modulation efficiency requires more subcarriers, thus consuming more transponder pairs.

Expectedly, the 1+1 Protection scheme is overwhelmingly expensive, frequently doubling the hardware cost (e.g., jumping from ~1260 to ~1872 in G17 M4). Establishing a redundant path inherently means doubling the transceivers required at the source and destination nodes.

6 Experiment Results Analysis

The benchmark and Noc-aware algorithm are experimented on the Itay 10-node and Germany 17-node network. Each network has M1, M2, M3, M4, and M5 traffic request matrices to meet requirements. The difficulty level increases with the number of traffic request matrices. The hardest level is M5. The results show that the Custom NoC-aware Best-Fit algorithm consistently reduces spectrum fragmentation compared with the Benchmark algorithm in both G17 and IT10 networks. In all traffic matrices, the total Number of Cuts (NoC) produced by the custom algorithm is lower, indicating that the spectrum is allocated in a more contiguous method, and fragmentation is effectively controlled.

In the G17 network, the custom algorithm also improves blocking probability under most traffic conditions. The improvement becomes more noticeable as traffic load increases. For example, in matrix M4 and M5 the blocking probability of the custom method is lower than the benchmark, suggesting that reduced fragmentation helps preserve larger continuous spectrum blocks for later requests.

In the IT10 network, the advantage of the custom algorithm is even more evident. While the benchmark algorithm begins to experience blocking matrices M4 and M5, the custom algorithm maintains very low blocking probability, demonstrating better resource utilization and allocation efficiency.

Overall, the results indicate that incorporating fragmentation awareness through NoC minimization improves spectrum organization and can reduce blocking probability, especially under heavier traffic loads. However, this benefit comes at the cost of higher computational complexity due to the need to evaluate multiple paths and candidate slot allocations.

7 Extended Analysis

7.1 constraint relaxation

In addition to comparing the algorithms, it is important to analyze the impact of the spectrum continuity and spectrum contiguity constraints.

7.1.1 Dropping Spectrum Continuity

You are assuming that the network nodes are equipped with ideal Wavelength/Spectrum Converters. A signal can arrive at a node on Frequency Slot Unit (FSU) 10 and leave on FSU 50.

Impact on Parameters of Interest:

Blocking Probability (BP%): Decreases significantly. Connections are no longer blocked just because a single common FSU block cannot be found end-to-end. If there is a gap on Link A and a different gap on Link B, the demand can be served.

Average Path Length: Decreases. In your current Custom Algorithm, demands often take longer detours ($K=2, 3$, etc.) just to find a path where the same slots are free all the way through. Without this constraint, the algorithm can stick to the absolute shortest physical paths.

Highest Used FSU & Average Link Load: Decreases. Spectrum can be packed much more densely because you are patching together mismatched gaps, meaning you don't have to push assignments into higher, unused frequency bands.

Spectrum Fragmentation (NoC): The negative impact of fragmentation disappears, but the raw NoC metric might actually increase. Because the algorithm is actively filling in random "potholes" of spectrum across different links, the visual fragmentation of the spectrum grid might look messier, even though it's no longer causing blocking.

Hardware Cost: Astronomically increases in the real world. While your transponder cost at the end nodes might remain stable, dropping this constraint requires installing highly expensive Optical-Electrical-Optical (O-E-O) converters or all-optical spectrum converters at every intermediate routing node.

7.1.2 Dropping Spectrum Contiguity

You are assuming a single demand can be split into non-adjacent spectrum blocks on the same link (e.g., a demand needing 6 FSUs can be transmitted using 2 FSUs at index 10, and 4 FSUs at index 100).

Impact on Parameters of Interest:

Blocking Probability (BP%): Decreases drastically. This completely neutralizes the fragmentation problem. Even if the network is highly fragmented, as long as the sum of the scattered free slots equals the required slots, the demand is accepted.

Spectrum Fragmentation (NoC): Becomes completely irrelevant as a performance bottleneck. The raw NoC value would artificially spike because you are actively dividing single demands into tiny, non-contiguous sub-bands.

Highest Used FSU: Reaches its absolute theoretical minimum. The algorithm will act like liquid, perfectly filling every single gap at the bottom of the spectrum grid before ever moving to higher frequencies.

Average Path Length: Decreases. Similar to dropping continuity, there is no need to take longer routes to find a massive, contiguous 10-slot gap. You can just take the shortest path and use the fragmented gaps.

Transponder Cost: Increases drastically. To transmit a single demand over non-contiguous spectrum bands, you need multiple independent laser sources/transceivers (one for each sub-band), or highly advanced multi-band transmitters. Furthermore, real-world physics would require you to insert "Guard Bands" between every single non-contiguous block, which actually wastes total spectrum capacity.

7.1.3 Dropping Both Constraints

You can use any free FSU on any link, regardless of whether they align end-to-end or sit next to each other.

Impact on Parameters of Interest:

The Network acts as a simple "Pipe": The complex RMSA problem devolves into a simple "Maximum Flow" mathematical problem.

BP% & Highest FSU: Both will drop to the mathematical absolute minimum possible for the given topology and traffic matrix. Blocking will only occur if the sheer volume of traffic strictly exceeds the total physical Gbps capacity of the fiber links (i.e., you run out of all 320 slots entirely).

Average Path Length: Will revert entirely to the fixed shortest path ($K=1$).

Cost: While it offers perfect spectrum utilization, the hardware required to achieve this (converters at every node + multiple fractional transponders per connection) makes it financially impossible to build in reality.

7.2 Path Protection

1+1 dedicated path protection ensures network reliability by establishing two link-disjoint paths for each traffic request such as a primary path and a backup path. The same data is transmitted simultaneously on both paths, allowing immediate switching to the backup if a failure occurs.

To extend the benchmark and custom algorithms, routing must first find a pair of link-disjoint paths. Spectrum slots are then allocated on both paths. The benchmark algorithm uses the shortest disjoint path pair with first-fit allocation, while the custom algorithm evaluates candidate path pairs and slot positions and selects the one that minimizes Noc.

Protection significantly increases network resource consumption, since spectrum must be reserved on two paths instead of one. As a result, used spectrum and blocking probability increase, and fragmentation may also rise. The results show that enabling 1+1 protection leads to much higher slot usage and blocking compared to the unprotected case, illustrating the trade-off between network reliability and resource efficiency.

7.3 Comparison to the twining group

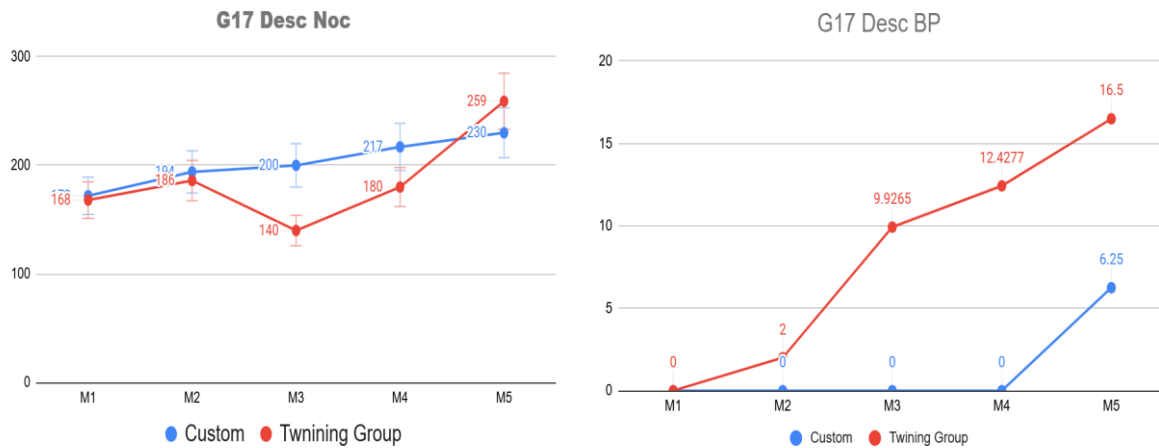


Fig 5. Comparison on G17

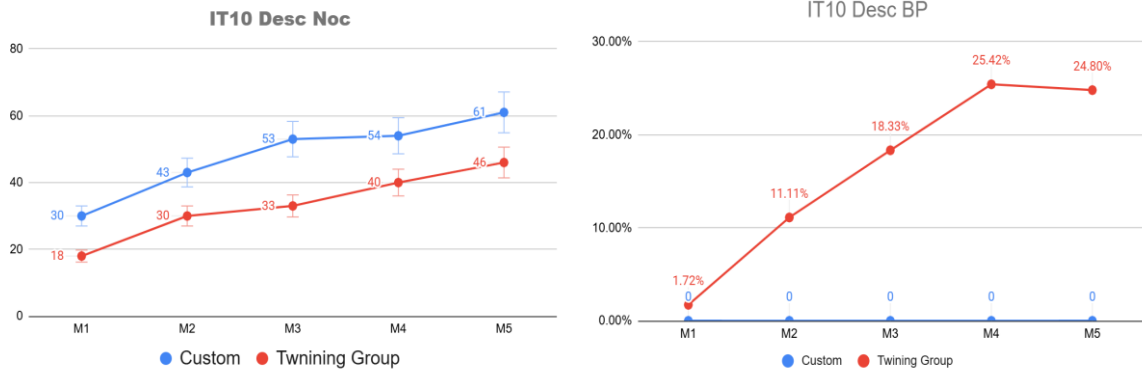


Fig 6. Comparison on IT10

Before comparing the Number of Connections (NoC), a fundamental prerequisite must be established: both systems must be evaluated under conditions that provide comparable Quality of Service (QoS). In network design, while minimizing NoC (representing the operator's infrastructure cost) is the primary objective, it must be strictly constrained by the Blocking Probability (BP, representing the user experience). Optimizing NoC becomes practically meaningless if the algorithm sacrifices fundamental service reliability. > Therefore, we prioritize evaluating BP to ensure a fair and realistic baseline. By verifying that both systems serve comparable traffic loads with similar blocking probabilities, we ensure the control of variables, allowing for a scientifically valid and meaningful comparison of their NoC efficiency.

Since the objective is to evaluate the customized algorithms, we will compare them under their optimal conditions. Specifically, we will use the results from the descending traffic sorting order, which yields the best performance for our designed algorithm. Furthermore, to provide a comprehensive evaluation, we will analyze the NoC performance across both complex networks (e.g., G17) and simpler, sparser topologies (e.g., IT10).

In most cases, the other group achieved lower NoC than ours. This is likely because our Best-Fit algorithm, while ensuring higher service rates, tends to create spectrum fragmentation.

However, we observed that their approach resulted in significantly higher blocking probability. This suggests a fundamental trade-off: our higher NoC may be the cost of serving a larger proportion of requests.

One possible explanation for the BP difference is the routing strategy. In sparse topologies like IT10, shortest-path algorithms have limited flexibility—when the primary path is blocked, finding alternative edge-disjoint paths becomes challenging, potentially leading to increased blocking.

Therefore, while direct NoC comparison shows their algorithm as more efficient, the performance picture changes when considering the served traffic load. A complete evaluation should consider both

NoC and BP together.

7.4 Conclusion

We implemented a benchmark algorithm based on first-fit and proposed a NoC-aware algorithm based on best-first. The algorithms have been analyzed on two networks to demonstrate our proposed scheme outperforms the benchmark algorithm. Besides, we also analyze the impact of spectrum constraints and path protection.

8 Extended Analysis

8.1 Statement of contributions

Congying Wang: Proposed and implemented the benchmark algorithm, converted the twining group result under the same metric.

Yongbo Liu: Proposed and implemented the NoC algorithm, analyzed the impact of spectrum algorithm.

Xuchen Zhu Paddington: Implemented and reviewed the benchmark algorithm, path protection, drafted the report, presentation work.

In general, everyone contributes equally and collaborates tightly within group to run this project smoothly. Good job to our dear teammates! (:

8.2 Reflection on group work

Congying Wang: I think our workflow could be improved. At the beginning, we should have a kick-off meeting to carefully read the assignment document together, list the key requirements, and align our understanding so that no details are missed later in the project. We should also clearly divide the coding tasks among team members to avoid overlapping work. In addition, when collaborating with the twinning group, we should communicate early about how the results will be compared, including agreeing on a communication platform, using the same standards to calculate parameters, and unifying the data file formats to make the final comparison straightforward.

Yongbo Liu: We have learnt so much about collaborations of teaming up not only within our group, but also outside group such as twining group. When it comes to something very puzzled, we should have voiced out and exchange our idea between twining group as it would have clarified each side's confusion.

Xuchen Zhu Paddington: Personally, I think our team did a great job as per this project. We proposed and executed algorithms to minimize the goal assigned in a limited time. I have nothing to say but a good job for my teammates.